

Periodic Geometry and Fixed Counts for Cyclic Rule 184

Route-A structural certificate C175

Abstract

For every ring length and every conserved particle number, we classify the periodic states of cyclic Rule 184. Below half density they are exactly the binary words without adjacent particles; above half density they are exactly the words without adjacent holes. The dynamics is a rotation on this core, which yields a closed fixed-count formula for every iterate.

Keywords: Rule 184; traffic cellular automaton; periodic point; cyclic independent set; Artin–Mazur zeta.

1 Periodic-core theorem

On

$$X_{N,k} = \{x \in \{0,1\}^{\mathbb{Z}/N\mathbb{Z}} : \sum_i x_i = k\}$$

let

$$F(x)_i = x_{i-1}(1 - x_i) + x_i x_{i+1}.$$

This is the simultaneous traffic move $10 \mapsto 01$. Put $m = \min(k, N - k)$ and let $(\rho x)_i = x_{i-1}$. If $k \leq N/2$, the periodic points are exactly the words with no cyclic 11, and $F = \rho$ there. If $k \geq N/2$, the periodic points are exactly the words with no cyclic 00, and $F = \rho^{-1}$ there. Thus every least period divides N .

At half filling the two descriptions agree and leave exactly the two alternating states, both of period two. For $k = 0$ and $k = N$, the unique uniform state is fixed.

2 Every-iterate fixed count

Let

$$I(g, r) = \begin{cases} 1, & r = 0, \\ \frac{g}{g-r} \binom{g-r}{r}, & 1 \leq r \leq \lfloor g/2 \rfloor, \\ 0, & \text{otherwise.} \end{cases}$$

This is the number of cyclic length- g binary words with r isolated marked sites. For $n \geq 1$, set $g = \gcd(N, n)$ and $q = N/g$. Then

$$\# \text{Fix}(F^n | X_{N,k}) = \begin{cases} I(g, m/q), & q \mid m, \\ 0, & q \nmid m. \end{cases}$$

An F^n -fixed word lies in the periodic core. There F^n is rotation by n in one of the two directions. Rotation invariance makes the word q repetitions of a cyclic block of length g ; the block contains m/q isolated minority symbols, proving the formula.

3 Route-A boundary

The periodic cycles are intrinsic but have no proved arithmetic semantics. The exact source formula is not a target divisor comparison. Accordingly the preliminary v0.2 tuple is

(AO_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),

with overall ROUTE_A_REJECTED and Route B false. No target table, arithmetic local factor, Euler factor, root number, automorphy input, or Hilbert–Pólya claim is used. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.